Decisions, October 24

The team settled the conflicts between the EMG plan and the earlier README, CLAUDE.md and HARDWARE.md. This page records what was decided and what replaced what. The plan that follows from it is PLAN.md.

#	Topic	Decision	Replaces
1	Who it's for and who demos	Laryngectomy patients are the audience. Austin, who is healthy, is the demo speaker; no patients take part in the demo. Performance and proof of concept come first, to impress the judges	"Healthy speaker only" (EMG plan) vs "patients" (README): both now hold, for different roles
2	Electrode sites	Gaddy sites 1, 2, 3, 5, 6, 8. No throat pad	Site 4 on the throat (EMG plan)
3	Reference pads	Each board has its own REF pad behind the ears, as in HARDWARE.md	One shared reference (EMG plan)
4	Modes	One mode: camera + EMG + Claude together, for free speech in real time. No separate camera mode or EMG mode. No closed-mouth speech, and no EMG training for it	Free speech camera-only plus a separate quick-phrases EMG mode (README); lips-sealed EMG mode (EMG plan)
5	Mouth	Open mouth, silent, camera always on	Lips sealed (EMG plan); camera off for quick phrases (README)
6	Fixing mistakes	Claude through the Claude API, for the Anthropic track. Claude follows the whole conversation, including the partner's words from Cartesia Ink. No local LLM	Local Qwen-class model (EMG plan)
7	When the voice speaks	Cartesia speaks in real time while Austin mouths, without waiting for the end of the sentence	Speaker confirms each sentence before playback (README, sponsor notes)
8	What gets trained	Only the EMG model (SVR-1), plus the decoder's weights fitted on Austin's recordings. Auto-AVSR is used as is; Claude is prompted	"No training" for free speech (README)
9	Phrase picking	Quick phrases are dropped, so the 10-phrase classifier and the Jev context check go with them. Claude covers context	Classifier top 3 + Jev (README)
10	Recording	About 75 minutes of open-mouth silent mouthing of varied sentences, with camera and EMG on one clock. No 50-repetition phrases	~2.5 h with closed-mouth sets (EMG plan); ~30 min of phrases (README)
11	Compute	Everything runs on Austin's laptop (RTX 5060, 115 W): camera, EMG, decoder, Claude calls, Cartesia, dashboard	Thai's Mac reading the pads, linked to Austin's laptop over a hotspot (HARDWARE.md)
12	Streaming lip reading	Evan builds it, following EVAN_WORKFLOW.md, which was written with the whole system in mind	Evan to design it alone
13	The demo's headline	Real-time free speech: live captions of both speakers, and Austin's cloned voice speaking while he mouths, with his facial emotion	Free speech plus quick phrases with the camera off (README); fused speed + closed-lips switch (EMG plan)

New decisions

- **Emotion:** the camera reads Austin's facial expression (Evan), and Cartesia applies it to the voice through `generation_config` (Frank).
- **The partner's speech:** a microphone records the partner, Cartesia Ink transcribes it into the captions, and Claude reads it as context.
- **Speaker control:** "confirm before playback" is gone (decision 7). Instead, a deliberate jaw clench, read by the EMG jaw sites, cancels the voice mid-sentence, and Claude answers "please repeat" when it cannot tell what was said. This keeps the responsibility story for the Cartesia judges.
- **SVR-1 changes:**
  - dropped: the fixed-phrase head with its "none" class, closed-lips transfer and the local LLM;
  - added: a speaking-or-not output, partial-sentence training, pace and effort for the voice, and the jaw-clench detector.

Details: ARCHITECTURE.md. - **Cartesia:** Sonic 3.6, with speed, volume and emotion through `generation_config`, per Cartesia's hackathon rules in `sponsors/cartesia.md`.

Still open

- **Electrical safety with one laptop:** the pads now connect to the laptop that charges. Either a galvanically isolated USB link between the Arduino and the laptop, or the laptop runs on battery whenever pads are on skin (the GPU then slows down). See HARDWARE.md. Thai decides, before pads go on skin.
- **Owners** of the Claude predict/repair service (proposed: Frank) and the captions dashboard (proposed: Thai).
- **Claude model:** `claude-opus-5-5` at low effort is the default; `claude-haiku-5-5` if measured repair latency is too high.